

OBESITY AND METABOLIC SYNDROME

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LEARNING OBJECTIVES

At the end of this learning session, students should be able to:

1. Discuss normal weight maintenance and define BMI.
2. Define obesity and define metabolic syndrome and their clinical abnormalities and causes.
3. Describe the relationship of overnutrition and metabolic consequences in various tissues.
4. Describe the roles and regulation of various hormones involved in appetite regulation, obesity, and metabolic syndrome including insulin, glucagon, leptin, ghrelin, PYY, CCK, plus others.
5. Describe Prader-Willi syndrome.
6. Identify the mechanisms of insulin resistance and how it develops.
7. Describe the metabolic dysfunctions present in metabolic syndrome and diabetes mellitus type II.

OUTLINE:

1. Regulation of body weight
2. Obesity
3. Fat deposition and body types
4. Epidemiology of obesity
5. Hormonal control of appetite
6. Clinical Correlations
 - Prader-Willi Syndrome
 - METABOLIC SYNDROME (big section)

BODY WEIGHT REGULATION AND THE BMI

$$\text{BMI} = (\text{weight in kg})/(\text{height in meters})^2$$

$$= (\text{weight in lb})/(\text{height in inches})^2 \times 703$$

Body Mass Index (BMI) correlate with the amount of body fat in most individuals:

Severely underweight <16

Underweight <18.5

Normal: 18.5 – 24.9

Overweight: 25 – 29.9

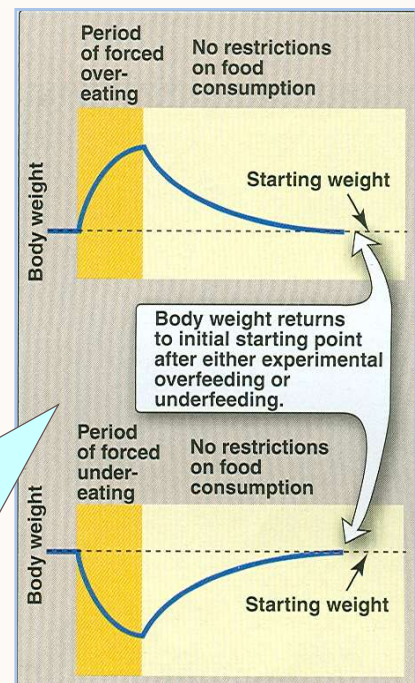
Obese ≥ 30

Severely obese ≥ 40

(Source: NIH and WHO Guidelines)

Each individual has a biologically predetermined 'set point' for body weight.

The body tends to add adipose tissue when the body weight falls below the set point and to lose weight when it falls below the set point





OBESEITY

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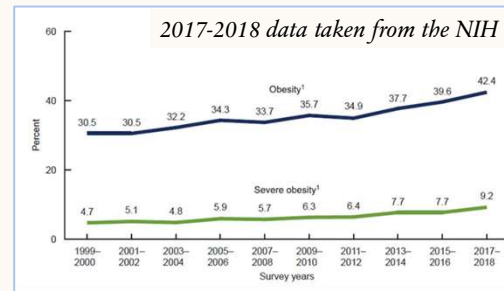
Disorder of body weight regulatory systems characterized by an **accumulation of excess body fat**.

Multifactorial Disorder:

1. Polygenic
2. Environmental

Other factors:

1. Neurologic
2. Psychological



Obesity in the US from 2001-2018

Habitual **excessive caloric intake over energy expenditure** can lead to obesity. Current abundance of food has encouraged North Americans to eat more while having a sedentary lifestyle. There is an **obesity epidemic** in the US with childhood obesity now becoming a major problem.

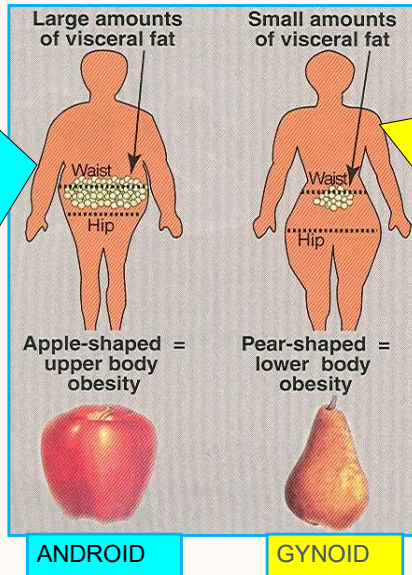


FAT DEPOSITION

Excess fat in the **central abdominal area**.

Associated with a greater risk for hypertension, insulin resistance, diabetes, dyslipidemia and coronary heart disease.

Waist to hip ratio:
> 0.8 for women
➤ >1.0 for men.

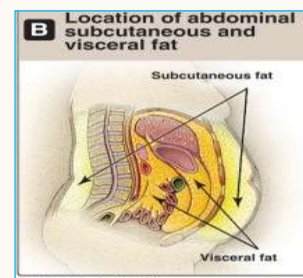


Fat in the **lower extremities** around the hips or gluteal region.

Relatively benign healthwise and is **commonly found in females**.

Waist to hip ratio: < 0.8 for women, <1.0 for men.

Generally, about 80-90% of fat stores are subcutaneous with the remainder being visceral (e.g., the intestines)



The Android or apple body shape has more **visceral fat** and poses a higher risk.

The gynoid or pear body shape has relatively **more subcutaneous fat** and poses less risk.

Viscera are the internal organs in the main cavities of the body, especially those in the abdomen,

EPIDEMIOLOGY OF OBESITY

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Figure 1. Prevalence of obesity in adults age 20 and older, by sex and age: United States, August 2021–August 2023

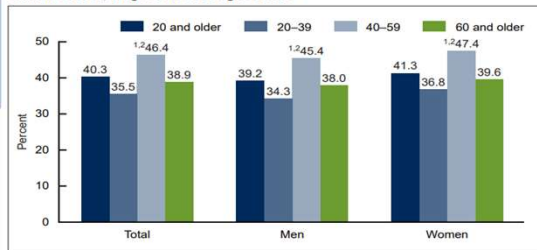
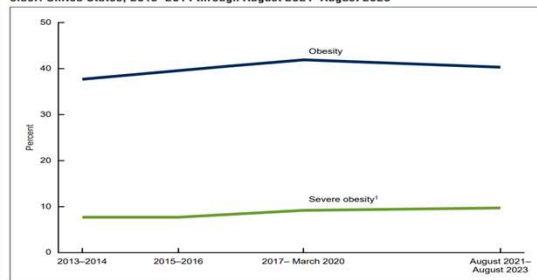


Figure 4. Trends in age-adjusted obesity and severe obesity prevalence in adults age 20 and older: United States, 2013–2014 through August 2021–August 2023



- BMI typically increases up to age 65 but declines after age 65 due to loss of muscle but also fat.
- All-cause mortality
 - ↑ proportionally to extent of obesity
 - ↑ proportionally to length of obesity < age 65
 - ↓ after age 65 for overweight (BMI 25-30) compared to normal weight (BMI 20-25)
 - poor health reduces appetite
 - age-related weight loss

Figures taken from the NIH/NIDDK website..

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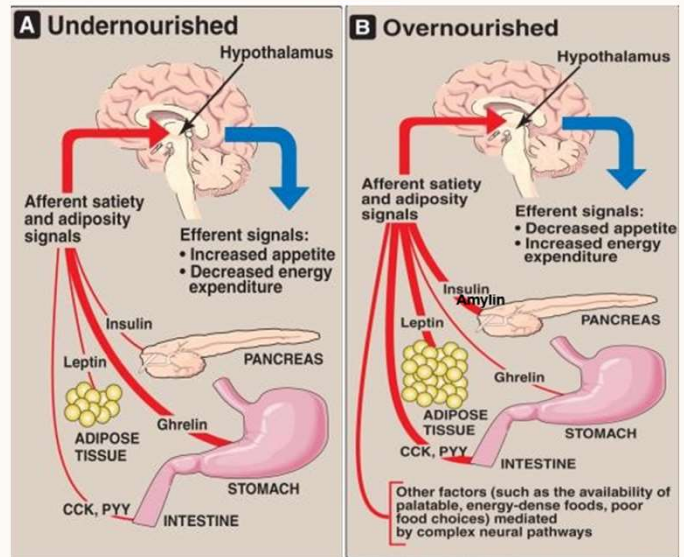
HORMONAL CONTROL OF APPETITE

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- Appetite stimulator:
 - Ghrelin (stomach)
- Appetite suppressants:
 - Insulin
 - Leptin
 - PYY
 - CCK – via the vagus nerve
 - Amylin
- Arcuate nucleus regulates appetite using NPY neurons and AgRP neurons.
- Regulated *via* melanocortin (MSH)

NPY – Neuropeptide Y

AgRP – Agouti-related Peptide



- 1) Appetite stimulants suppress melanocortin (alpha-MSH or MSH).
- 2) Melanocortin inhibits appetite
- 3) Melanocortin is stimulated by appetite suppressants: CCK, PYY, Leptin, Insulin, Amylin
- 4) Proopiomelanocortin, POMC neurons are stimulated by leptin and insulin to reduce feeding behaviors and reduce appetite.
 - a) Alpha MSH down regulates appetite
 - b) Alpha MSH also regulates skin color and hair color.
- 5) Arcuate nucleus contains neurons that make both NPY and AgRP.
 - a) NPY and AgRP have overlapping effects on feeding and energy balance.
 - b) Agouti related peptide, AgRP. Inhibits the actions of melanocortins, thus suppressing leptin action.
 - c) Neuropeptide Y is a hormone that stimulate foraging and eating behaviors.
 - i. Over expression causes hyperphagia.
 - ii. NPY neurons are inhibited by PYY, Amylin, insulin and leptin and thus act as appetite suppressors.
 - iii. Ghrelin induces NPY transcription thereby increasing appetite

CCK – cholecystokinin

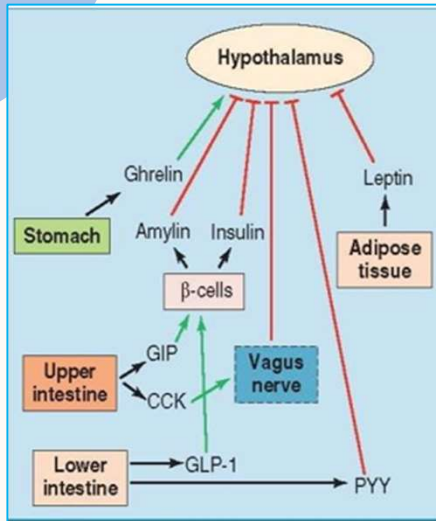
PYY – Peptide YY or peptide tyrosine tyrosine; binds to NPY receptors

GLP1 – Glucagon-like peptide 1.

AgRP – Agouti-Related Peptide

GHRELIN: APPETITE STIMULANT

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- Produced in the stomach - gastric mucosa
- Release regulated by nutrients (not by distention)
- Released during fasting
- Increases appetite and food intake
- Decreases fat utilization
- **Activates NPY/AgRP neurons** in arcuate nucleus in hypothalamus which are appetite stimulating hormones
- **Up regulates NPY receptor gene (↑↑transcription)**
- Overproduction of ghrelin leads to obesity
- **Ghrelin levels tend to be lower in obese people**

Ghrelin is a 28-amino acid peptide from the oxyntic glands in the stomach.

Functions:

Released during short-term fasting, suppressed after meal. This depends on nutrients, not distension of stomach.

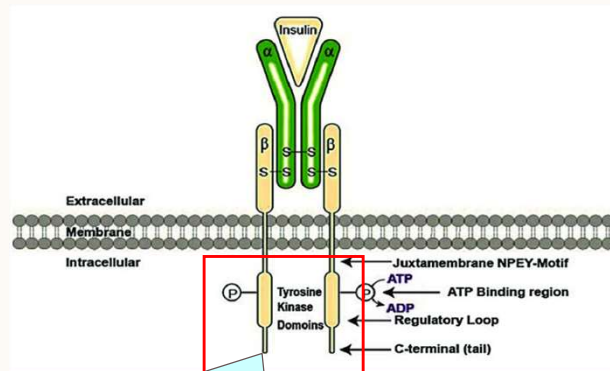
Induces feeding by stimulating neurons in the arcuate nucleus that express neuropeptide Y and agouti related protein (AgRP).



INSULIN: APPETITE SUPPRESSANT

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- **Insulin central effects:**
- **Stimulates POMC/CART cells**
 - Increases alpha-MSH
 - Satiety
 - Increases energy expenditure
- **Inhibits NPY/AgRP cells**
 - Decreases appetite



The **tyrosine kinase domain** of the insulin receptor gets activated once bound to insulin.

POMC = Proopiomelanocortin

CART = Cocaine and amphetamine regulated transcript.

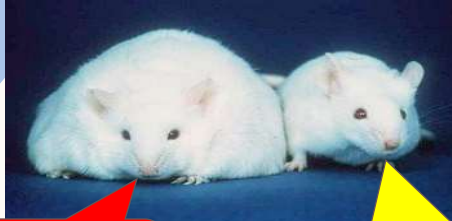
NPY = Neuropeptide Y

AgRP= Agouti related protein

Stimulation of POMC/CART cells increases alpha-MSH (melanocortin), which suppresses appetite.

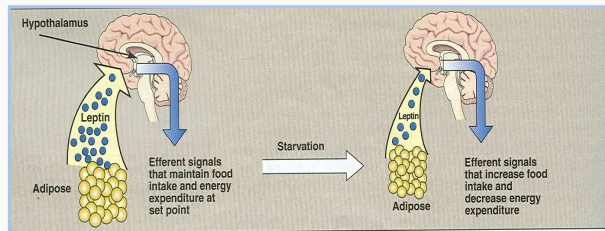
LEPTIN: APPETITE SUPPRESSANT

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I have a mutated Ob gene!

I have a normal Ob gene!



- Coded by the *ob gene*; secreted by the **adipocytes**
- Receptors on hypothalamus via 2 second messenger pathways:
 - **JAK/Stat** down regulates endocannabinoids
 - **MAPK** up-regulates growth and development
- **↓ NPY production and secretion**
- **↓ AgRP release**
- **↑ insulin sensitivity**
- Activates POMC neurons
 - Signals “satiety” by **stimulating POMC** release; Up regulates Luteinizing Hormone Releasing Hormone (LHRH) secretion
- Modulates T-cell receptors and is angiogenic
- **Leptin stimulates TRH secretion!**

POMC = Pro-opiomelanocortin

CART = Cocaine and amphetamine regulated transcript



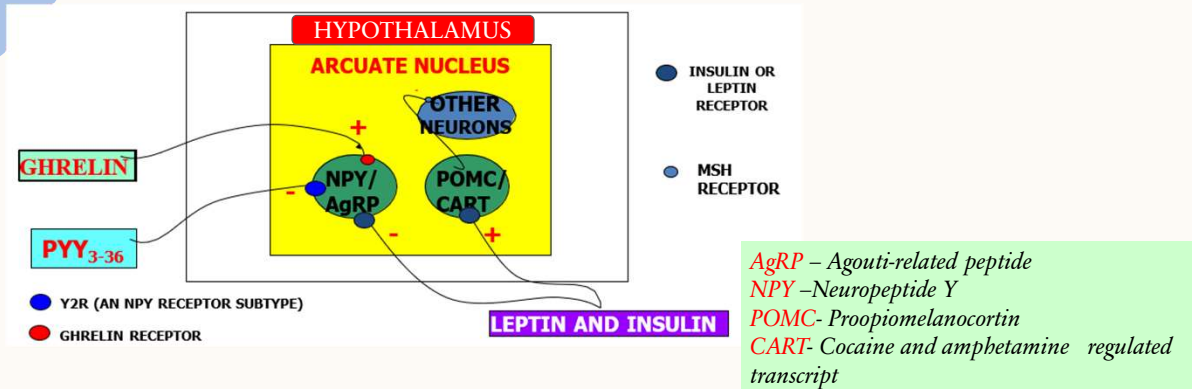
PYY: APPETITE SUPPRESSANT

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- A peptide secreted by the ileum and colon
- Increases in proportion to caloric intake
 - Leads to decrease in food intake
- Actions in arcuate nucleus:
 - Inhibits NPY/AgRP neurons
 - Stimulates POMC/CART cells
 - POMC release
 - POMC processing in hypothalamus → release of alpha-MSH
 - alpha-MSH → inhibit food intake; increased energy use
- ↑CART → inhibit food intake; increased energy use

APPETITE CONTROL IN THE HYPOTHALAMIC LEVEL

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Afferent signals (proteins) that affect the arcuate nucleus in the hypothalamus.

Summary of appetite control at the Hypothalamic level

Arcuate nucleus is the appetite control center in hypothalamus. Two important cell types: (Secrete neuropeptides)

NPY/AgRP (Neuropeptide Y/Agouti Related Peptide)

POMC/CART (Pro-Opiomelanocortin/Cocaine and amphetamine-regulated transcript)

NPY and AgRP: Stimulate appetite; Inhibit energy expenditure

POMC converted to α -MSH

CART and α -MSH: Inhibit food intake; Stimulate energy expenditure

Leptin and insulin

Stimulate POMC/CART neurons $\rightarrow$ $\uparrow$ CART and α -MSH

Inhibit NPY/AgRP neurons $\rightarrow$ $\downarrow$ NPY & AgRP

Net effects: Satiety and appetite

Ghrelin

Stimulates NPY/AgRP $\rightarrow$ $\uparrow$ NPY AND AgRP Secretion

Net effect: $\uparrow$ appetite

PYY₃₋₃₆ is a homologue of NPY

Binds to an inhibitory receptor on cells containing NPY/AgRP

$\downarrow$ Secretion of NPY and AgRP

Net effect: $\downarrow$ appetite

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PRADER-WILLI SYNDROME (PWS)

Imprinting disorder in which **ghrelin** level is elevated.

–Paternal chromosome 15 q arm deletion (70%) or maternal uniparental dysomy (25%)

–Symptoms:

- hypotonia
- failure to thrive/poor sucking reflex

–Early childhood onset of **hyperphagia**

–Severe obesity-related cardiovascular problems

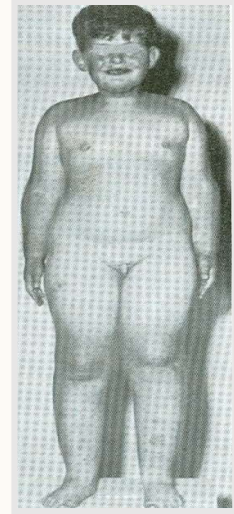
–**Metabolic syndrome** developing into Type II Diabetes

–**Hypogonadism**

–Short stature; small hands and feet

–Intellectual impairment

In PWS, the **absence of Magel2 gene** disrupts the normal functioning of POMC neurons, leading to issues with satiety and appetite regulation



Prader-Willi syndrome (PWS) is a **genetic multisystem disorder** characterized during infancy by lethargy, diminished muscle tone (hypotonia), a weak suck and feeding difficulties with poor weight gain and growth and other hormone deficiency.

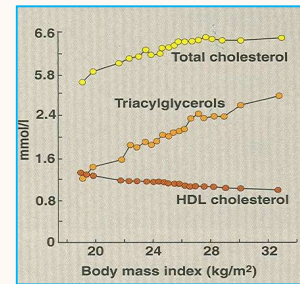
Watch the movie to understand the chromosomal effects. hyperphagia is **extreme hunger and feeding**

METABOLIC SYNDROME

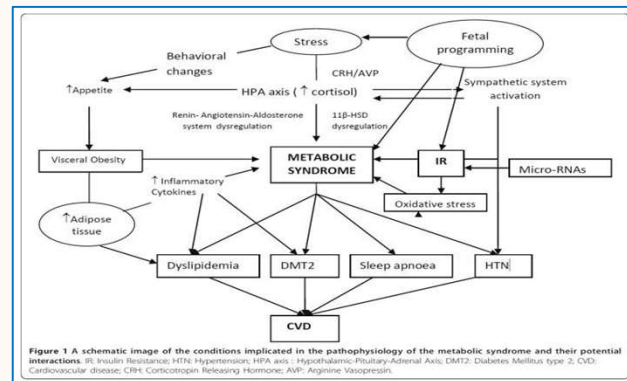
- A cluster of abnormalities:
 - Abdominal obesity
 - Hyperglycemia; Insulin resistance; Hyperinsulinemia
 - Dyslipidemia ($\downarrow$ HDL $\uparrow$ LDL $\uparrow$ TAG $\uparrow$ cholesterol)
 - Hypertension

- Conditions that may develop as a result of metabolic syndrome:

Polycystic ovary syndrome (PCOS)
 Non-alcoholic fatty liver, Gallstones
 Asthma, Cancer, Sleep problems
 Cardiovascular problems, Diabetes



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In metabolic syndrome, there is an increased risk for developing diabetes mellitus and cardiovascular disorders.

Progression of Metabolic Syndrome:

- Leads to:
 - Dyslipidemia
 - Diabetes
 - Sleep apnea
 - Cardiovascular disease

Dyslipidemia is a **high level of lipids** **Overview of Cholesterol and Lipid Disorders** The body needs fats for growth and energy. It also uses them to synthesize hormones and other substances needed for the body's activities.

Sleep apnea is a **serious sleep disorder that happens when a person's breathing is interrupted during sleep**. People with untreated sleep apnea stop breathing repeatedly during their sleep, sometimes hundreds of times during the night.



METABOLIC SYNDROME CRITERIA

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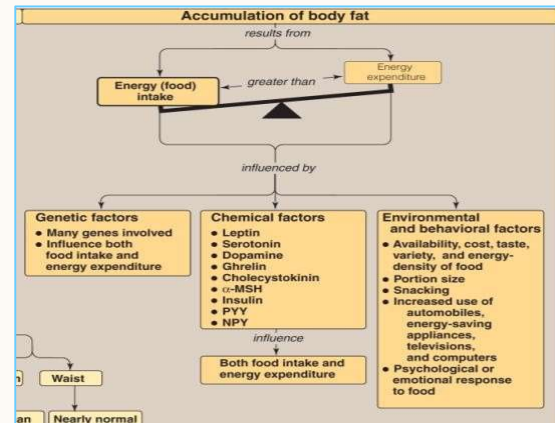
According to the US National Heart, Lung and Blood Institute (NHLBI):

- **Abdominal obesity:**
Waist circumference ≥ 35 in males; ≥ 40 in females
- **High blood pressure** $\geq 130/80$ mm Hg
- **Impaired fasting blood glucose** ≥ 100 mg/dL
- **High triglyceride** ≥ 150 mg/dL
- **Low HDL** ≤ 40 mg/dL in men; ≤ 50 mg/dL in females

PROGRESSION TO METABOLIC SYNDROME

- Metabolic syndrome progression is linked to the following:

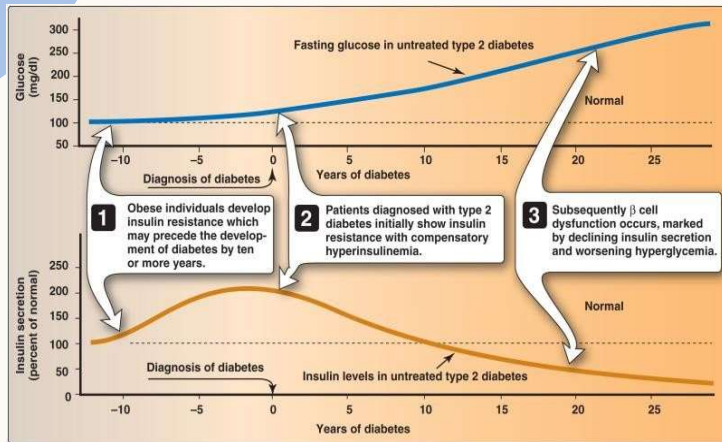
- Poor dietary choices
- Lack of physical activity
- Epigenetic changes during fetal development and SNPs in defined genes
- Disruption of the Hypothalamic-Pituitary Axis
- Metabolic and inflammatory dysregulation



Gestational diabetes is **a type of diabetes that can develop during pregnancy in women who don't already have diabetes**. Every year, 2% to 10% of pregnancies in the United States are affected by gestational diabetes. Managing gestational diabetes will help make sure patient has a healthy pregnancy and a healthy baby.

METABOLIC SYNDROME AND INSULIN RESISTANCE LEADING TO TYPE 2 DM

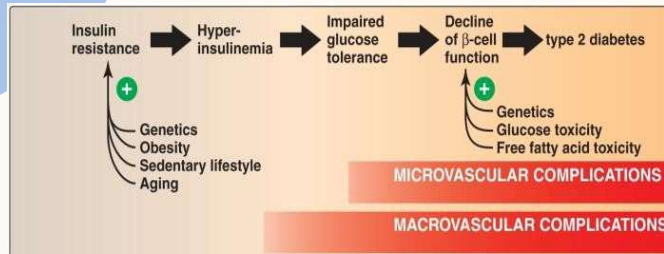
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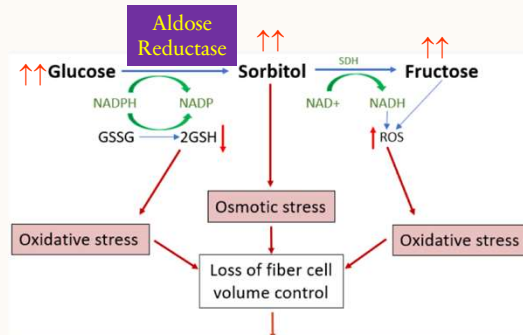
Acanthosis nigricans is associated with insulin resistance

In **insulin resistance**, glucose control is maintained initially but a higher level of insulin is needed. Obesity is linked to insulin resistance.

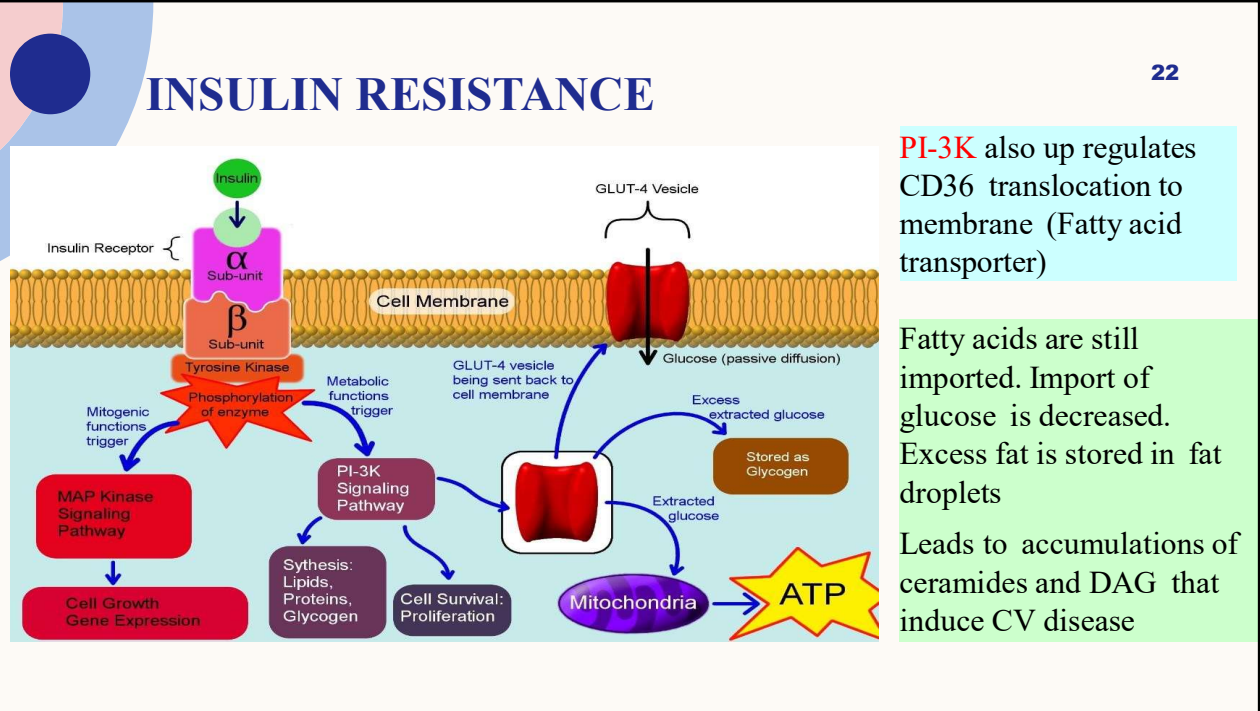
PROGRESSION OF METABOLIC SYNDROME



GLUT4 translocation is impaired in insulin resistance leading to significantly decreased uptake of glucose in the muscles and the adipocytes. Metformin increases GLUT4 expression and translocation to the plasma membrane.



Microvascular Complications of DM
 Retinopathy/cataracts
 Nephropathy
 Neuropathy



- 1) Mitochondrial numbers are decreased in obesity This is the cause of the lower metabolic activity.
- 2) Some insulin signaling remains the same
 - a) IRS 1 via PI3K continues to mobilize CD36 to the plasma membrane.
 - b) Since the Insulin levels are increased a higher fraction of the CD36 is always on the cell surface in obesity, metabolic syndrome and diabetes.
 - c) Increased fatty acid uptake with less β oxidation (due to fewer mitochondria) results in pooling of fatty acids and DAGs into fat droplets in most cells.
- 3) Obese individuals have altered signaling via the PI-3K- Akt pathway.
 - a) Glut 4 movement is decreased so blood glucose remains high
- a) Glycogen, lipid and protein synthesis is slowed
- b) Growth and proliferation of cells is reduced, so short stature correlates with childhood obesity
- 4) Akt-2 is decreased in obesity
 - a) Akt-2 knockout mice are a model for type 2 diabetes.
 - b) Akt-2 knock out mice have impaired resistance to tumorigenesis.

CD36 is a scavenger receptor that functions in **high affinity tissue uptake of long chain fatty acids (FA)** and contributes under excessive fat supply to lipid accumulation and metabolic dysfunction.



PROGRESSION OF METABOLIC SYNDROME: INCREASE IN ADIPOKINES

- **Leptin** – decreases hunger but has low penetrance across blood brain barrier in obesity, leading to **leptin resistance**
- **Apelin** regulates
 - insulin secretion
 - acid secretion in stomach
 - NO release in endothelia
- **Monocyte chemotactic protein-1** (MCP-1, CCL2)
 - induces **amylin** (glycemic regulator)
 - Induces insulin resistance
- **Retinol binding protein 4** (RBP4)
 - induces insulin resistance,
 - signals adipocytes when glucose is low
- **Chemerin** – regulates lipolysis
- **Interleukin-6** (IL-6) – stimulates inflammation
- **Plasminogen activator inhibitor-1** (PAI-1) activate inflammation, fibrosis and thrombosis in vessels
- **Resistin**
 - Indicator for insulin resistance and inflammation

Amylin is a peptide hormone that is co-secreted with insulin from the pancreatic β - cell and is thus deficient in diabetic people. It inhibits glucagon secretion, delays gastric emptying, and acts as a **satiety agent**.

Amylin replacement could therefore possibly improve glycemic control in some people with diabetes.



PROGRESSION OF METABOLIC SYNDROME: DECREASE IN ADIPONECTIN

Adiponectin actions:

- ↑ insulin sensitivity
- ↑ Glycolysis and β -oxidation
- ↓ Gluconeogenesis
- ↓ Adipocyte differentiation
- ↑ Peroxisomal fatty acid oxidation
- Uncoupling TCA cycle and electron transport from fatty acid degradation
- ↓ TNF α
- Synergistic to leptin
- Antagonistic to ghrelin

In metabolic Syndrome:

- ↓↓Adiponectin
- ↑inflammation
- ↑↑ Insulin resistance
- Leptin resistance

Note:

In non-obese persons, adiponectin and leptin increase and decrease together.

But NOT IN Obese persons, Metabolic Syndrome, and T2 Diabetes

- 1) The role of adiponectin in preventing some diseases is poorly understood.
- 2) Action on endothelium of blood vessels may be due to blockage of oxidative stress pathways
- 3) Action on hypothalamus is not well understood.
- 4) Blood levels inversely related to BMI.

PROGRESSION OF METABOLIC SYNDROME: HORMONE CHANGES

1) **Insulin resistance** ↓energy use ↑↑GLUCOSE

- a) ↓Glycogenesis
- b) ↑Lipolysis

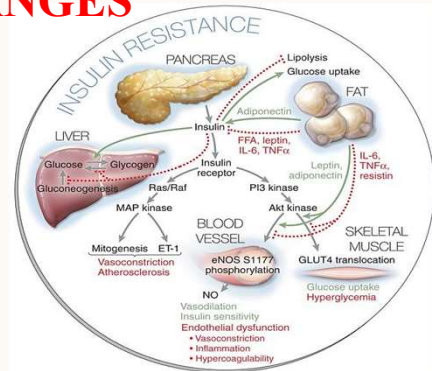
2) **Adipokine secretions altered:**

- a) ↓Adiponectin leads to ↓ insulin sensitivity
- b) ↑Leptin, free fatty acids, IL-6 and TNF alpha

IL-6, TNF-alpha and **resistin** inhibit Akt pathways in muscle and endothelium

↑Oxidative stress

↓GLUT4 translocation



Insulin resistance:

↓PI3K-Akt Pathway

↑NF-κβ (↑inflammation)

↑JAK/STAT (↑inflammation)

- 1) Green arrows and red dotted lines are the normal pathways.
- 2) The red text is the effects of obesity/metabolic syndrome on the pathways affected by insulin resistance.
- 3) Important note- these changes occur in obesity prior to insulin resistance and worsen with the onset of insulin resistance.
- 4) In disease state, Insulin has different effects.
 - a) Glut 4 is not mobilized to cell surface.
 - b) eNOS is not activated (decreases blood flow and increases atherosclerosis)
- 5) Fat cells alter their adipokine secretions
 - a) Adiponectin is decreased, reducing insulin secretion
 - b) Leptin, free fatty acids, IL-6 and TNF alpha inhibit Insulin secretion
 - c) IL-6, TNF alpha and resistin inhibit Akt pathways in muscle and endothelium
 - i. Endothelia become oxidatively stressed leading to hypertension, inflammation and atherosclerosis
 - ii. Muscles cells cannot translocate GLUT4 so, hyperglycemia occurs.

PROGRESSION OF METABOLIC SYNDROME: BODY FAT CHANGES

Insulin Resistance caused by:

- ↑ Resistin
- ↑ DAG
- ↑ free FA

Atherogenic dyslipidemia due to:

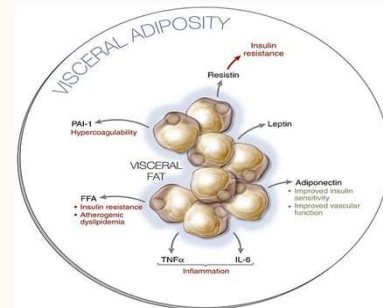
- ↑ DAG
- ↑ free FA

Vascular hypercoagulability due to ↑ PAI-1

Inflammation caused by:

- ↑ IL6
- ↑ TNF α

Waist to hip ratio INCREASES!



Insulin resistance:

↑↑ β -oxidation of FA without a corresponding increase in the TCA and ETC

1. Abdominal and especially visceral abdominal adipose tissue is associated with metabolic syndrome.
2. Obesity leads to increased numbers of macrophages in adipose tissue.
 - a. macrophages adopt a pro-inflammatory phenotype.
 - b. Macrophages secrete tumor necrosis factor α (TNF- α) and other cytokines.
 - c. Macrophage cytokines have paracrine effects on the adipocytes.
 - d. TNF- α activates Ser/Thr kinases that phosphorylate IRS-1, thereby uncoupling insulin signaling. (note this is the effect that decreases Glut 4 mobilization)

Atherogenic dyslipidemia (AD) refers to **elevated levels of triglycerides (TG) and small-dense low-density lipoprotein** and low levels of high-density lipoprotein cholesterol (HDL-C).

PROGRESSION OF METABOLIC SYNDROME: BLOOD PRESSURE CHANGES

PI3K-Akt pathway inhibition

- Caused by: $\uparrow$ IL-6; $\uparrow$ TNF α ; $\uparrow$ free FA
- Insulin resistance occurs

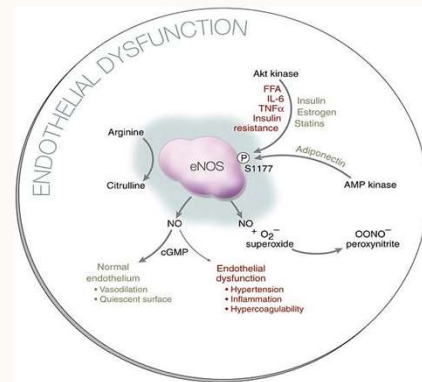
Endothelial function disrupted by eNOS-uncoupling from Akt $\rightarrow \downarrow\downarrow$ NO

$\uparrow$ blood pressure

$\uparrow$ inflammation

$\uparrow$ hypercoagulability

- Increased ROS production: $\uparrow$ oxidative stress



- 1) Akt kinase pathway is inhibited by free fatty acids, IL-6 TNF alpha and result in Insulin resistance in endothelial cells.
- 2) Nitric oxide production is inhibited, This becomes a disruptor of endothelial function and lack of vasodilation.
 - a) Hypertension
 - b) Inflammation
 - c) Hypercoagulability

PROGRESSION OF METABOLIC SYNDROME: LIPID CHANGES

↑ **Free FA and TAG** in liver

↑ VLDL

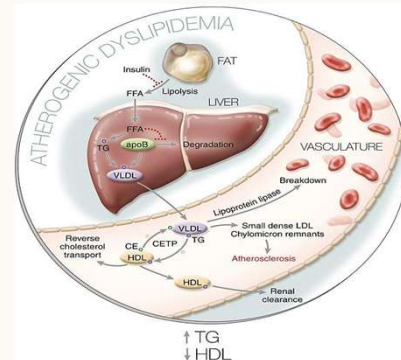
↑ TAG

↑ Chylomicrons

↑ **Plaque formation**

↑ small dense LDL

**HYPERLIPIDEMIA
ATHEROSCLEROSIS**



Insulin resistance:

↑↑ceramide (lipotoxic) → ↑leptin resistance
↓insulin sensitivity

↑ LDL (“bad cholesterol”) favored by high caloric intake, especially carbohydrate and saturated fat.

↓ HDL (“good cholesterol”), decreased by lack of exercise.

Dietary effects

High carbohydrate diet:

High triglyceride, high LDL cholesterol, low HDL cholesterol.

Carbohydrate replaced by fat:

All fats reduce VLDL triglycerides moderately.

All fats raise HDL, saturated $\geq$ monounsaturated $>$ polyunsaturated.

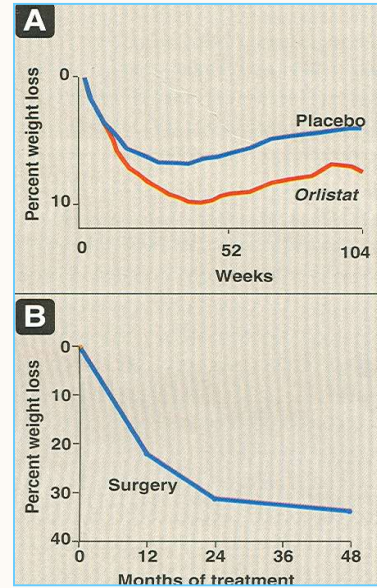
LDL is reduced substantially by polyunsaturated fat, slightly by monounsaturated and not at all by saturated fat.

Fish oil reduces triglycerides.

DO YOU WANT TO LOSE WEIGHT?

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- ↑↑ Physical activity: Let's MOVE!
- Caloric restriction – diet
- Examples of drugs used for weight reduction
 1. **Sibutramine**- appetite suppressant that inhibits reuptake of both serotonin and norepinephrine
 2. **Orlistat** – lipase inhibitor
 3. **Semaglutide** –glucagon-like-peptide 1(GLP-1) receptor agonist
- Surgical treatment





Encircle the change in the given compound/process/ enzyme observed in a 45-year-old man suffering from metabolic syndrome compared with a normal individual.

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Compound/Process	Encircle		
Insulin	↑	↓	no change
HbA1c	↑	↓	no change
PYY	↑	↓	no change
Ghrelin	↑	↓	no change
Adiponectin	↑	↓	no change
POMC	↑	↓	no change

Compound/Process	Encircle		
Leptin	↑	↓	no change
Fibrinolysis	↑	↓	no change
NF- κ B Pathway	↑	↓	no change
Oxidative stress	↑	↓	no change
Brown adipocyte activity	↑	↓	no change
LDL/HDL	↑	↓	no change



SUMMARY

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- Metabolic syndrome is also associated with:
 - Non-alcoholic fatty liver disease (NFLD)
 - Polycystic Ovary Syndrome (PCOS)
- Not all obese persons develop metabolic syndrome
- Not all persons with metabolic syndrome are obese
- Complex interactions between dietary habits, hormonal status and genetic background
- Association between poor nutrition during fetal development and increased risk of adult cardiovascular disease

Reflect on your thinking and learning today:

- ⚙ Which part of the learning session did you encounter struggles today and how do we deal with it?
- ⚙ Which part of the learning session brought you the most satisfaction?



THANK YOU

*Live as if you were to die tomorrow.
Learn as if you were to live forever.
- Mahatma Gandhi*

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 Keyword Chart 1 34	
Keyword	Description/property/clinical correlate
Ghrelin	
PI3K-Akt Pathway	
Ob gene	
POMC	
Prader-Willi Syndrome	
Acanthosis nigricans	

 Keyword Chart 2 35	
Keyword	Description/property/clinical correlate
CCK	
Adiponectin	
Android body type	
Agouti-related peptide (AgRP)	
Aldose reductase	
PYY	



Practice Makes Perfect!

36

A 29-year-old woman has a height of 5 feet and 2 inches; her waist measures 40 inches and her hip 35 inches. Over the past 10 years, she had been on nine different diets for a period of 3 weeks losing around 10 pounds. On discontinuation of each diet, she regained weight returning to 200 pounds. This woman will most likely have which of the following?

- A. Low levels of resistin.
- B. High levels of adiponectin
- C. An android pattern of fat distribution.
- D. Low levels of circulating triacylglycerol.
- E. A BMI which indicates that she is moderately overweight



Practice Makes Perfect!

37

A 51-year-old man, who stands 5 feet and 8 inches, has a waist circumference of 40 inches and weighs 150 kg. He has a blood pressure of 160/95 mm Hg and a fasting blood glucose of 175 mg/dL. He Which of the following is a lipoprotein profile consistent with this patient?

	Chylomicrons	VLDL
A	Increased	No significant change
B	Increased	Increased
C	Decreased	Increased
D	Increased	Decreased
E	No significant change	Increased



Practice Makes Perfect!

38

A 29-year-old woman, who is on the obese side, wanted to lose weight drastically by daily aerobic exercise and strict low-carbohydrate and low-fat diet. What will happen to the adipocytes of this woman when she loses weight?

	Number of adipocytes	Size of Adipocytes
A	↓	↓
B	↑	↑
C	No significant change	↓
D	↓	No significant change
E	No significant change	No significant change



Practice Makes Perfect!

39

A 25-year-old woman, who wanted to lose weight drastically without much physical effort, was advised to take sibutramide. What is the mechanism of action of this drug?

- A. Inhibiting pancreatic lipase
- B. Stimulating reuptake of serotonin and epinephrine
- C. Inhibiting reuptake of serotonin and epinephrine
- D. Stimulating gastric and pancreatic lipase
- E. Blocking action of cholecystokinin



Practice Makes Perfect!

40

To follow progress of treatment with intensive insulin on a 45-year-old diabetic man, the physician ordered the lab to determine the level of glycated/glycosylated hemoglobin (Hb_{A1C}). Hb_{A1C} has which of the following properties?

- A. The α -chain is the point of attachment for glucose in Hb_{A1C} .
- B. Hb_{A1C} allows monitoring of glucose over a longer period of time.
- C. Hydroxyurea enhances the formation of Hb_{A1C} .
- D. The formation of Hb_{A1C} predisposes diabetics to a hemolytic crisis.
- E. Hb_{A1C} level goes up upon intensive treatment with insulin.